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app.py
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from flask import Flask, request, jsonify
from flask_cors import CORS
import os
from werkzeug.utils import secure_filename
from detection import detect_fake_news, detect_ai_media

app = Flask(__name__)
CORS(app)

UPLOAD_FOLDER = 'uploads'
os.makedirs(UPLOAD_FOLDER, exist_ok=True)
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER

@app.route('/')
def index():
    return "Backend is running"

@app.route('/detect-news', methods=['POST'])
def detect_news():
    data = request.get_json()
    text = data.get('text', '')
    if not text:
        return jsonify({'error': 'No text provided'}), 400
    result = detect_fake_news(text)
    return jsonify({'result': result})

@app.route('/detect-media', methods=['POST'])
def detect_media():
    if 'file' not in request.files:
        return jsonify({'error': 'No file uploaded'}), 400
    file = request.files['file']
    filename = secure_filename(file.filename)
    filepath = os.path.join(app.config['UPLOAD_FOLDER'], filename)
    file.save(filepath)
    result = detect_ai_media(filepath)
    os.remove(filepath)
    return jsonify({'result': result})

if __name__ == '__main__':
    app.run(debug=True)

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detection.py
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import joblib
import os
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from PIL import Image
import numpy as np

MODEL_PATH = "fake_news_model.pkl"
VEC_PATH = "vectorizer.pkl"

if not os.path.exists(MODEL_PATH) or not os.path.exists(VEC_PATH):
    data = {
        "text": [
            "Aliens invaded Earth!",
            "Apple launched new iPhone.",
            "Vaccines turn people into zombies.",
            "NASA confirms flat Earth theory.",
            "Water is essential for life."
        ],
        "label": [1, 0, 1, 1, 0]
    }
    df = pd.DataFrame(data)
    vectorizer = TfidfVectorizer()
    X = vectorizer.fit_transform(df["text"])
    y = df["label"]
    model = LogisticRegression()
    model.fit(X, y)
    joblib.dump(model, MODEL_PATH)
    joblib.dump(vectorizer, VEC_PATH)
else:
    model = joblib.load(MODEL_PATH)
    vectorizer = joblib.load(VEC_PATH)

def detect_fake_news(text):
    x_test = vectorizer.transform([text])
    prediction = model.predict(x_test)[0]
    confidence = model.predict_proba(x_test).max()
    label = "Fake News" if prediction == 1 else "Real News"
    return f"{label} (Confidence: {confidence*100:.2f}%)"

def detect_ai_media(filepath):
    if filepath.lower().endswith(('.jpg', '.jpeg', '.png')):
        try:
            img = Image.open(filepath).convert('RGB')
            img = img.resize((128, 128))
            img_np = np.array(img)
            variance = np.var(img_np)
            if variance < 100:
                return "Possibly AI-Generated Image"
            else:
                return "Likely Real Image"

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except:
    return "Error processing image"
elif filepath.lower().endswith(('.mp4', '.avi', '.mov')):
    return "Video detection not implemented"
else:
    return "Unsupported file type"
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requirements.txt

Flask

flask-cors

scikit-learn

pandas

numpy

Pillow

joblib

gunicorn

Procfile

web: gunicorn app:app